SIMATS ENGINEERING

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA12	**Course Title:**	Computer Architecture
CO Assessed:	CO1 – Precision (BL4)	**Assessment:**	Assessment Tool 1 – Application based questions
Weightage:	40%	**Total Marks:**	100
CO1 Statement:	Analyse the functional units of a computer system including CPU, memory, bus structures, and I/O components by examining instruction execution cycles, addressing modes, and performance metrics such as CPI, clock cycles, and benchmarking		
Student Name:	ASWIN. M	**Reg. No.:**	192412437
Section / Batch:	2024 3rd year (ECE)	**Date:**	15/07/2026

Instructions to Students

- Identify the relevant computer architecture concepts, functional units, or performance parameters required to solve the given application-based problem.
- Analyse the interaction between CPU, memory, bus structures, and I/O components in the given scenario.
- Apply appropriate instruction execution cycles, addressing modes, and architectural concepts to determine the correct solution.
- Compute performance metrics such as CPI, clock cycles, CPU execution time, speedup, or benchmarking measures using appropriate formulas and calculations.
- Interpret the calculated results and explain their impact on overall computer system performance.

QUESTIONS

1. A web developer notices that the displayed colour does not match the expected output because an incorrect hexadecimal colour code was entered. To identify the source of the mismatch, use an online colour picker or graphics software to compare two hexadecimal colour values (for example, #FF0000 and #F00000). Analyse how changing a hexadecimal value affects the binary representation and the displayed output with suitable illustration.
2. A programmer notices that a program produces different outputs when different input values are used. To examine the effect of varying input values, execute an 8085 program that performs arithmetic operations using different input values. Observe the changes in the Sign, Zero, Carry, and Parity flags after each operation. Analyse how the flag register influences program execution and determines why different inputs produce different flag values.
3. A user notices that a computer responds slowly while editing a document, browsing the internet, and playing audio simultaneously. To analyse the system behaviour, perform these activities together and observe the system's response. Analyse how the CPU, memory, and I/O devices cooperate to execute these tasks, and identify which functional unit is primarily responsible for processing, storing, and receiving data during the operations.

Code of Conduct

I ASWIN.M - 192412437 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: Aswin. M

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Q. No	Understanding of Problem Context (20)	Application of Concepts (30)	Problem-Solving Approach (20)	Accuracy of Solution (20)	Clarity (10)	Total (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 20	/ 30	/ 20	/ 20	/ 10	/ 100

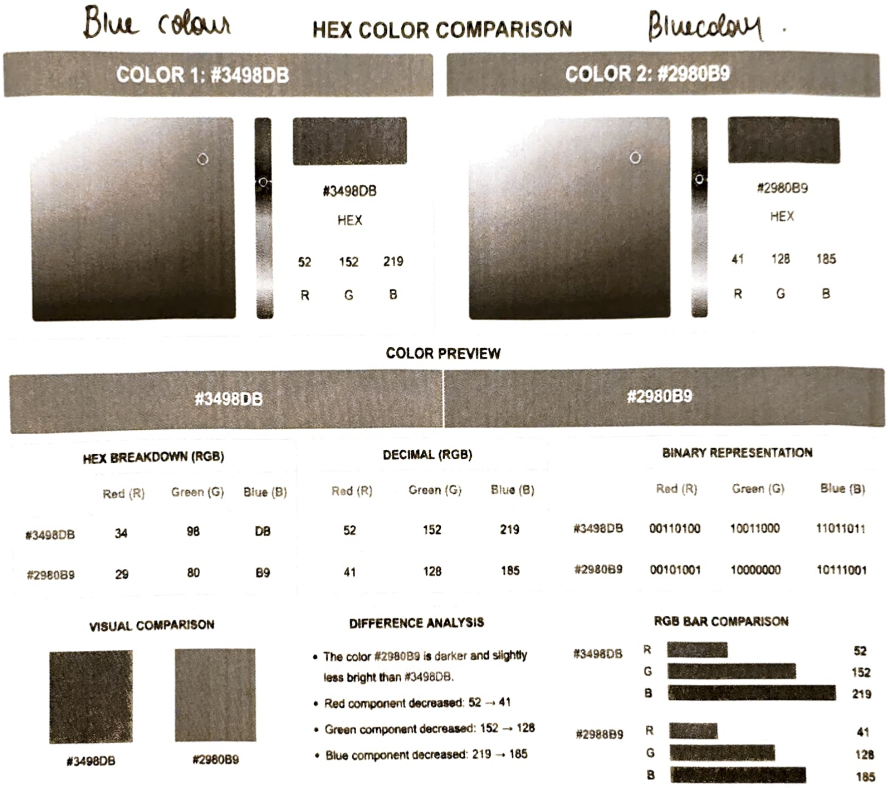
Course Coordinator **Faculty Incharge**

CO1 ASSESSMENT TOOL - 1

Aswin M - 192412437
CSA1220 - Computer Architecture

1. A web developer notices that the displayed colour does not match the expected output because an incorrect hexadecimal colour code was entered. To identify the source of the mismatch, use an online colour picker or graphics software to compare two hexadecimal colour values (for example, #FF0000 and #F00000).
Analyse how changing a hexadecimal value affects the binary representation and the displayed output withsuitable illustration.

Blue colour Bluecolour.



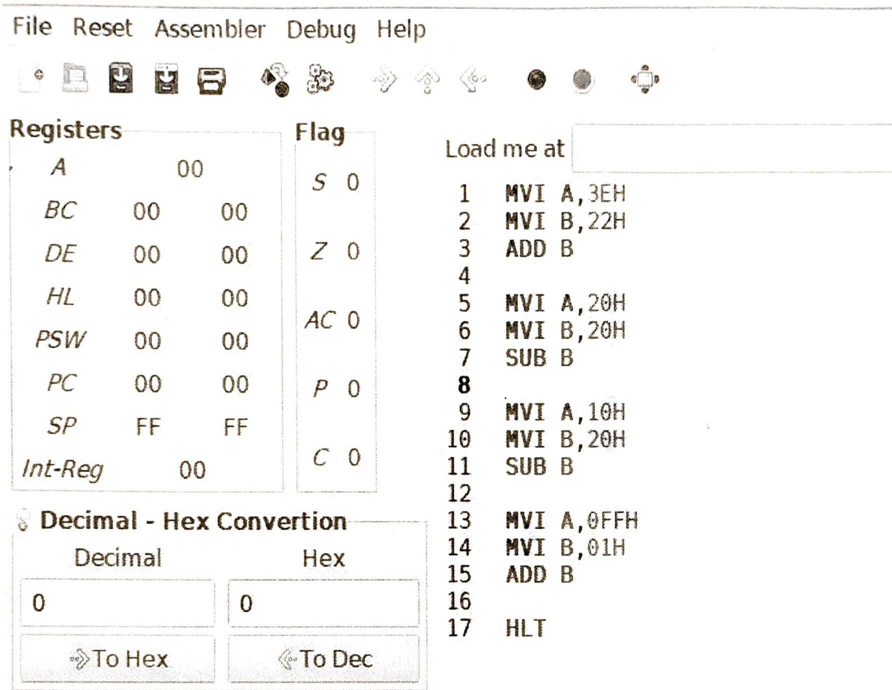
Colour 1 : Hex-Code Value is #3498DB
Colour 2 : Hex-Coded Value is #2980B9

→ Two hexadecimal colour codes were compared using a colour picker

→ A small change in the hexadecimal value altered the RGB values and binary representation.

2. A programmer notices that a program produces different outputs when different input values are used. To examine the effect of varying input values, execute an 8085 program that performs arithmetic operations using different input values. Observe the changes in the Sign, Zero, Carry, and Parity flags after each operation. Analyse how the flag register influences program execution and determines why different inputs produce different flag values.

Before Execution



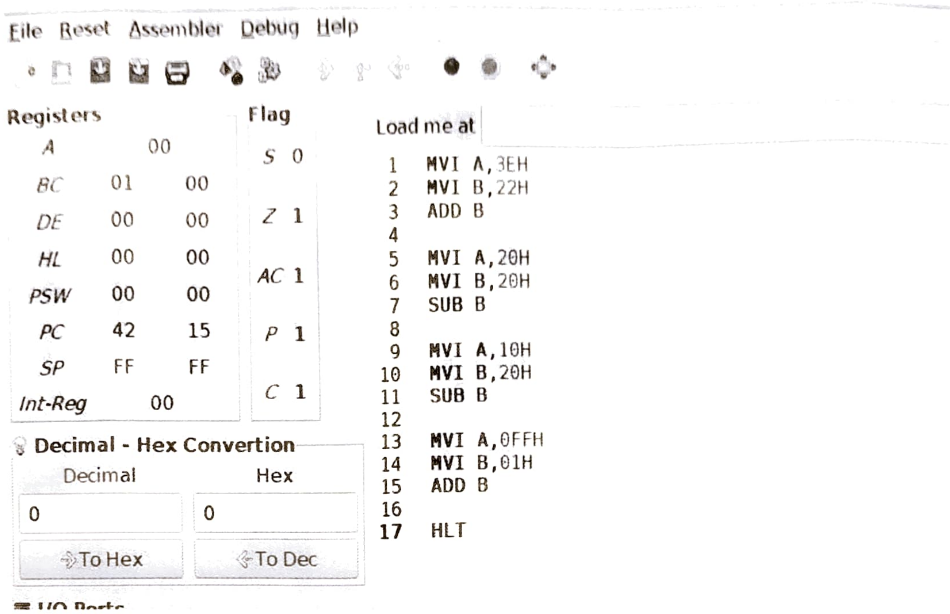
→ The arithmetic program was executed with different I/p values.

→ Different I/p produced different flag combinations.

→ After each ADD and SUB
The sign Z, C, P, S flags change according to the I/p.

After Execution

GNUSim8085

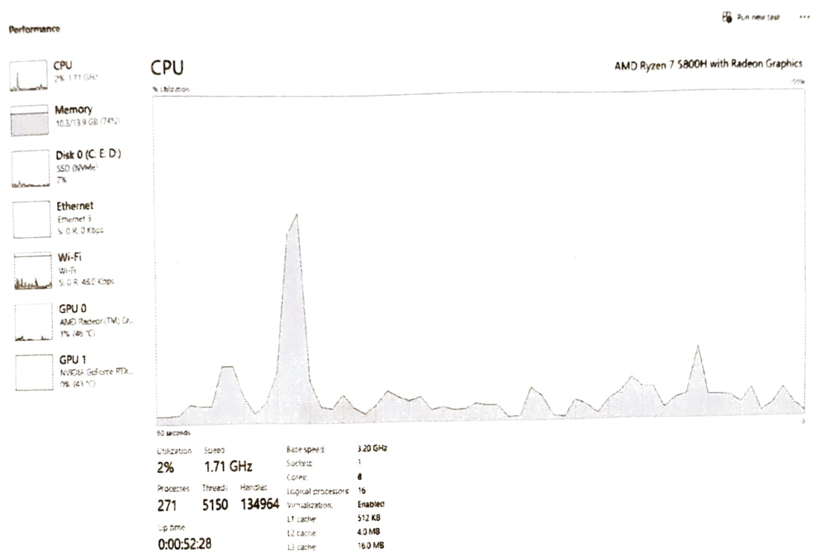


Register/Flag	Value
A	00H
B	01H
S	0
Z	1
CY	1
P	1

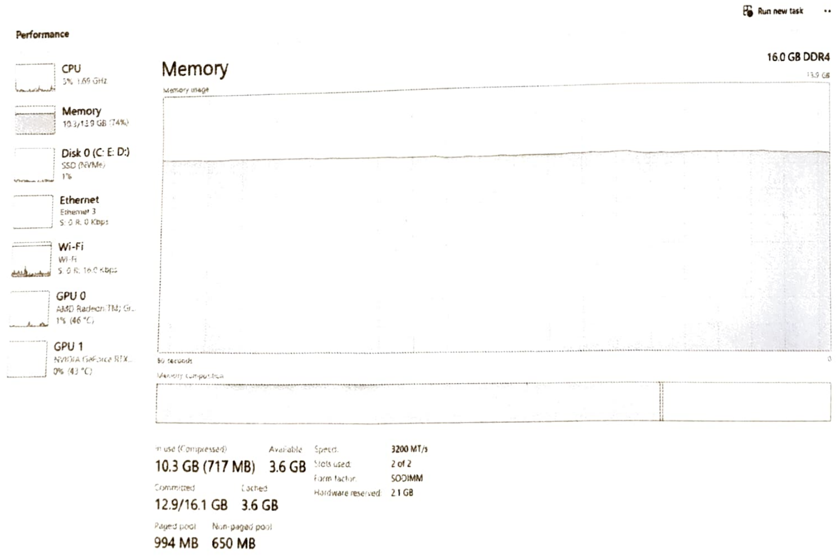
→ This flag register helps the processor make decisions during program execution.

3. A user notices that a computer responds slowly while editing a document, browsing the internet, and playing audio simultaneously. To analyse the system behaviour, perform these activities together and observe the system's response. Analyse how the CPU, memory, and I/O devices cooperate to execute these tasks, and identify which functional unit is primarily responsible for processing, storing, and receiving data during the operations.

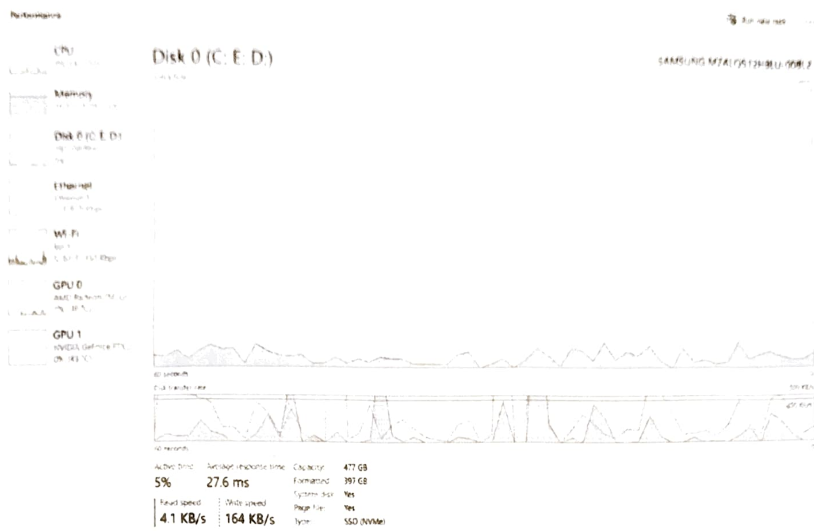
CPU Utilization



Memory Utilization



Disk Utilization



→ The CPU processed instructions from all running applications.

→ Memory temporarily stored program code and data required for execution.

→ Disk and I/O activity occurred continuously to access application files and data.